# Linear Algebra and Its Applications

## Section 4.S: Supplementary Exercises

Detailed worked solutions

Mathjock

**Text and numbering.** David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

**Source scope.** The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

**How to read the solutions.** Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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## 4.S Chapter 4 supplementary exercises

These exercises connect spanning sets, bases, coordinate changes, and rank. When proving two sets equal, establish both inclusions. When proving a universal claim false, give one concrete counterexample. For controllability, form $[\,B\ AB\ \cdots\ A^{n-1}B\,]$ and check whether its columns span all of the state space $\mathbb{R}^n$.

### Exercise 4.S.1: Twenty statements tying the chapter together

(a) **True.** The set is $\text{Span}\{v_1, \ldots, v_p\}$, a subspace of $V$ by the span theorem.

(b) **True.** Adding another generator cannot remove combinations already obtainable. A spanning subset remains spanning after a vector is added.

(c) **False.** Start with the independent list $\{e_1\}$ in $\mathbb{R}^2$ and add $e_1$ again; alternatively add $2e_1$. The enlarged list is dependent.

(d) **False.** In $\mathbb{R}^2$, $\{e_1\}$ is independent but does not span $\mathbb{R}^2$, so it is not a basis.

(e) **True.** The Spanning Set Theorem removes any redundant vectors from a finite spanning list, leaving an independent spanning subset.

(f) **True.** A list of exactly $\dim V = p$ vectors that spans $V$ is a basis by the Basis Theorem, hence is independent.

(g) **False.** A plane not through zero is not a subspace. For example the plane $z = 1$ misses the zero vector.

(h) **False.** In $A = [\,I_2\ I_2\,]$ the nonpivot columns are $e_1, e_2$, which are independent. Each is a combination of pivot columns, but that does not force a dependence among the nonpivot columns themselves.

(i) **True.** Rows $(1, 0), (1, 0)$ have the relation $r_2 - r_1 = 0$; replacing row two by row two minus row one changes the list to $(1, 0), (0, 0)$, which no longer has that same relation.

(j) **False.** Elementary row operations replace $Ax = 0$ by an equivalent system, so they preserve the null space.

(k) **False.** The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 0 \end{smallmatrix}\right)$ has two nonzero rows but rank one. Count nonzero rows only in an echelon form.

(l) **False.** Nullity is the number of columns minus rank, $n - k$, not generally $m - k$.

(m) **True.** Row-equivalent matrices have the same row space, so their row-space dimensions, and hence ranks, agree.

(n) **False.** Nonzero rows need not be independent, as the matrix in part (k) shows. Nonzero rows of an echelon form do give a row-space basis.

(o) **True.** Row operations preserve row spaces, and both row spaces equal that of the common reduced echelon form.

(p) **True.** Choose a basis $h_1, \ldots, h_d$ for $H$, where $d \leq 3$, and use these as columns of a $3 \times 3$ matrix, filling any remaining columns with zeros. Its column space is $H$. If $H = \{0\}$, use the zero matrix.

(q) **False.** Rank $m$ means onto, not necessarily one-to-one. For $A = [\,1\ 0\,]$, rank is $m = 1$ but $(0, 1)$ is a nonzero kernel vector.

(r) **True.** Onto $\mathbb{R}^m$ means the column space has dimension $m$, which is exactly rank $m$.

(s) **True.** The reverse change-of-coordinates matrix is the inverse, since changing from one basis to the other and back restores every vector.

(t) **False.** Column $j$ of $P_{C \leftarrow B}$ is $[b_j]_C$, not $[c_j]_B$. The latter columns occur in the inverse matrix.

### Exercise 4.S.2: A parametrization has a redundant generator

Collect the coefficients of $a, b, c$ to get generators

$$u = (1, 2, -1, 3), \quad v = (-2, 5, -4, 1), \quad w = (5, -8, 7, 1).$$

But $w = u - 2v$, so the third generator is redundant. The first two are independent: if $\alpha u + \beta v = 0$, the first two coordinates give $\alpha - 2\beta = 0$, $2\alpha + 5\beta = 0$, hence $9\beta = 0$ and both vanish.

**Conclusion.** A basis is $\{(1, 2, -1, 3), (-2, 5, -4, 1)\}$. There are three parameters in the original expression, but only two independent directions.

### Exercise 4.S.3: Describe a plane by a homogeneous equation

Both $u_1 = (-2, 4, -6)$ and $u_2 = (1, 2, -5)$ satisfy $x_1 + 2x_2 + x_3 = 0$. They are not multiples, so their span is a two-dimensional subspace contained in the plane defined by that equation. To verify equality without relying only on dimensions, let $b$ satisfy $b_1 + 2b_2 + b_3 = 0$. Solve the first two equations of $au_1 + cu_2 = b$: they give

$$a = (b_2 - 2b_1)/8, \qquad c = b_1 + 2a.$$

With these choices the third coordinate is $-6a - 5c = -b_1 - 2b_2 = b_3$.

**Conclusion.** $W = \{b \in \mathbb{R}^3 : b_1 + 2b_2 + b_3 = 0\}$.

### Exercise 4.S.4: A scalar multiplier cannot depend on the input variable

The equality $g(t) = tf(t)$ uses the function $t$, not one fixed real scalar. Linear dependence would require constants $a, b$, not both zero, with $a(3 + t) + b(3t + t^2) = 0$ for every $t$. Comparing the $t^2$ coefficient gives $b = 0$; comparing the constant gives $3a = 0$, so $a = 0$ as well. Thus the pair is independent, and the proposed discussion incorrectly treats a variable multiplier as a scalar.

### Exercise 4.S.5: Delete redundant polynomials from the given list

The polynomials are $p_1 = 1 + t$, $p_2 = 1 - t$, $p_3 = 4$, $p_4 = t + t^2$, $p_5 = 1 + 2t + t^2$. First $p_3 = 2p_1 + 2p_2$, so deleting $p_3$ does not alter the span. Next $p_5 = p_1 + p_4$, so delete $p_5$. The remaining $p_1, p_2, p_4$ are independent: in $ap_1 + bp_2 + cp_4 = 0$, the quadratic coefficient gives $c = 0$; then the constant and linear coefficients give $a + b = 0$ and $a - b = 0$, forcing $a = b = 0$.

**Conclusion.** $\{1+t, 1-t, t+t^2\}$ is a basis for $H$.

### Exercise 4.S.6: A basis of a known two-dimensional span

At least one of the four spanning polynomials is nonzero; choose it as $p$. Since their span has dimension two, at least one other polynomial $q$ is not a scalar multiple of $p$. Find such a $q$ by comparing coefficients. Two nonproportional nonzero vectors are independent, so $\{p, q\}$ is an independent pair in the two-dimensional space $H$. By the Basis Theorem it is already a basis. There is no need to row reduce all four coefficient vectors when the dimension is known in advance.

### Exercise 4.S.7: The homogeneous information that certifies every right-hand side

For eighteen equations in twenty variables, every nonhomogeneous right-hand side is attainable precisely when rank is eighteen. By rank–nullity this is equivalent to nullity $20 - 18 = 2$. Thus it suffices, and is necessary, to know that the homogeneous solution set is two-dimensional: exactly two independent homogeneous solutions span it.

### Exercise 4.S.8: Equal finite dimensions force equality of nested spaces

Let a basis of $H$ have $n$ vectors. Because $H \subseteq V$, these are $n$ independent vectors in the $n$-dimensional space $V$. The Basis Theorem says they span $V$. But their span is also $H$, so $H = V$. The containment hypothesis matters: distinct subspaces can have equal dimensions without either containing the other.

### Exercise 4.S.9: Rank and nullity of injective and surjective maps

**(a) If $T$ is one-to-one.**

Its kernel is zero, so rank–nullity gives $\dim \operatorname{range} T = n$. This also requires $n \leq m$, since the range is contained in $\mathbb{R}^m$.

**(b) If $T$ maps onto $\mathbb{R}^m$.**

Its range has dimension $m$. Thus its kernel has dimension $n - m$, which in particular requires $n \geq m$.

### Exercise 4.S.10: A maximal independent set must span

Suppose the independent set $S$ failed to span $V$. Choose $v \in V \setminus \operatorname{Span} S$. Then adjoining $v$ still leaves an independent set: if $av + \sum_i c_i s_i = 0$ is a finite relation and $a \neq 0$, it would express $v$ as a combination of elements of $S$, a contradiction. Thus $a = 0$, and independence of $S$ forces all $c_i = 0$. This contradicts maximality. Hence $S$ spans, and being independent as well, it is a basis. The argument covers possibly infinite $S$ because linear combinations always involve only finitely many terms.

### Exercise 4.S.11: A minimal finite spanning set is independent

Suppose a finite spanning set were dependent. Then a nontrivial relation would allow one vector with a nonzero coefficient to be solved for as a combination of the others. Deleting that vector would preserve every original linear combination, and thus preserve the span $V$. That contradicts minimality. Therefore the set is independent; since it already spans $V$, it is a basis.

### Exercise 4.S.12: Rank cannot increase beyond either factor's rank

**(a) Compare column spaces.**

Every output $ABx = A(Bx)$ belongs to $\operatorname{Col} A$, so $\operatorname{Col}(AB) \subseteq \operatorname{Col} A$. Taking dimensions gives $\operatorname{rank}(AB) \leq \operatorname{rank} A$.

**(b) Apply that result to the transposes in reversed order.**

Using $(AB)^T = B^T A^T$ and transpose invariance of rank,

$$\operatorname{rank}(AB) = \operatorname{rank}(B^T A^T) \leq \operatorname{rank} B^T = \operatorname{rank} B.$$

Both inequalities follow. Repeated application proves the corresponding bound for a product with more than two factors.

### Exercise 4.S.13: An invertible left factor preserves rank

The product inequality gives $\operatorname{rank}(PA) \leq \operatorname{rank} A$. Since $A = P^{-1}(PA)$, applying it again gives $\operatorname{rank} A \leq \operatorname{rank}(PA)$. The two opposite inequalities imply equality. Invertibility is what supplies the reverse factorization.

### Exercise 4.S.14: An invertible right factor preserves rank

The product inequality first gives $\operatorname{rank}(AQ) \leq \operatorname{rank} A$. The identity $A = (AQ)Q^{-1}$ gives the reverse inequality $\operatorname{rank} A \leq \operatorname{rank}(AQ)$, hence equality. Equivalently one can apply Exercise 4.S.13 to $(AQ)^T = Q^T A^T$.

### Exercise 4.S.15: A zero product limits two ranks

If $AB = 0$, every column of $B$ is in $\operatorname{Nul} A$. Thus $\operatorname{Col} B \subseteq \operatorname{Nul} A$, and dimensions give

$$\operatorname{rank} B \leq \dim \operatorname{Nul} A = n - \operatorname{rank} A.$$

Rearrangement yields $\operatorname{rank} A + \operatorname{rank} B \leq n$. The shared $n$ is the number of inputs of $A$ and outputs of $B$, which makes the containment meaningful.

### Exercise 4.S.16: Rank is subadditive

Take rank factorizations $A = C_A R_A$, $B = C_B R_B$, with ranks $r, s$. Then block multiplication gives

$$A + B = \begin{bmatrix} C_A & C_B \end{bmatrix} \begin{pmatrix} R_A \\ R_B \end{pmatrix}.$$

The left block matrix has $r+s$ columns and therefore rank at most $r+s$. By the product inequality, $\operatorname{rank}(A+B) \le r+s = \operatorname{rank} A + \operatorname{rank} B$. If either matrix has rank zero, it is zero and the inequality is immediate, so the proof also covers those degenerate cases.

## Exercise 4.S.17: Find an invertible square submatrix of full rank

### (a) Select independent columns.

Choose the $r$ pivot columns of $A$ to obtain an $m \times r$ matrix $A_1$. They are independent, so $\operatorname{rank} A_1 = r$.

### (b) Select independent rows of that submatrix.

Its row rank also equals $r$, so choose $r$ independent rows of $A_1$. The resulting $r \times r$ matrix $A_2$ has $r$ independent rows, hence rank $r$ and is invertible by the Invertible Matrix Theorem. This proves the requested existence; the case $r = 0$ has the usual empty square-matrix convention or may be excluded as trivial.

## Exercise 4.S.18: Expand four controlled time steps

### (a) Compute each state before collecting the blocks.

From $x_0 = 0$ and $x_{k+1} = Ax_k + Bu_k$,

$$\begin{aligned} x_1 &= Bu_0, \\ x_2 &= ABu_0 + Bu_1, \\ x_3 &= A^2Bu_0 + ABu_1 + Bu_2, \\ x_4 &= A^3Bu_0 + A^2Bu_1 + ABu_2 + Bu_3 \\ &= \underbrace{[\,B \;\; AB \;\; A^2B \;\; A^3B\,]}_{M} \begin{pmatrix} u_3 \\ u_2 \\ u_1 \\ u_0 \end{pmatrix}. \end{aligned}$$

The order of the input blocks is reversed to match the stated order in $M$. Each $u_j$ has two entries, so the stacked input has eight entries, matching the $4 \times 8$ matrix $M$.

### (b) Use full row rank to realize an arbitrary target.

Controllability means rank $M = 4$, so $M$ maps $\mathbb{R}^8$ onto $\mathbb{R}^4$. Given $v \in \mathbb{R}^4$, solve $Mz = v$, then split its eight entries into four two-entry blocks $(u_3, u_2, u_1, u_0)$. The computation in part (a) shows that this control sequence gives $x_4 = v$.

## Exercise 4.S.19: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B \;\; AB \;\; A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 0 & 1 & 0 \\ 1 & -\frac{9}{10} & \frac{81}{100} \\ 1 & \frac{1}{2} & \frac{1}{4} \end{pmatrix}, \qquad \det M = \frac{14}{25}.$$

Since this determinant is nonzero, $M$ has rank 3 and its columns span the state space. Therefore the pair $(A, B)$ is controllable.

### Exercise 4.S.20: Check the rank of the controllability matrix

Here the state dimension is 3 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & \frac{1}{2} & \frac{19}{100} \\ 1 & \frac{7}{10} & \frac{9}{20} \\ 0 & 0 & 0 \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so $M$ does not have full rank and $(A, B)$ is not controllable. Its actual rank is 2. Its last row is zero, while the leading $2 \times 2$ minor has determinant $1/5 \neq 0$, verifying rank two.

### Exercise 4.S.21: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{8}{5} \\ 0 & -1 & \frac{8}{5} & -\frac{24}{25} \\ -1 & \frac{8}{5} & -\frac{24}{25} & -\frac{3}{125} \end{pmatrix}, \qquad \det M = 0.$$

The determinant is zero, so $M$ does not have full rank and $(A, B)$ is not controllable. Its actual rank is 3. For example its leading $3 \times 3$ minor has determinant $-1$, showing rank at least three; the zero full determinant then gives exactly three.

### Exercise 4.S.22: Check the rank of the controllability matrix

Here the state dimension is 4 and the input has one coordinate, so form $M = [\,B\ AB\ A^2B\ A^3B\,]$. Exact arithmetic gives

$$M = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & -1 & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & \frac{229}{20} \\ -1 & \frac{1}{2} & \frac{229}{20} & -\frac{411}{40} \end{pmatrix}, \qquad \det M = -\frac{3}{10}.$$

Since this determinant is nonzero, $M$ has rank 4 and its columns span the state space. Therefore the pair $(A, B)$ is controllable.